

HUYNH TAN KHOA

Contact Information:

Email: khoadautay@gmail.com | Phone: 0905605271 | Location: Phu Nhuan district, Ho Chi Minh city
LinkedIn: linkedin.com/in/khoa-huynh-tan-34b505196 | GitHub: github.com/KhoaDauTay

EDUCATION

University of Greenwich (Vietnam) - Da Nang Campus

Bachelor of Computing | 2019 - 2023 | First class graduation

- Top 3 Student (Fall 2019, Spring 2020)
- Leader of Greenwich's Club of Developers (2021-2022)

PROFESSIONAL EXPERIENCE

Spartan

Backend AI Engineer | 04/2023 - 06/2026

Led development of AI-powered startup products involving RAG systems, LLM orchestration, and scalable backend infrastructure

- Designed and built the company-wide Python project template adopted across all subsequent AI projects, covering FastAPI project structure, reusable RAG pipeline base classes, OpenAI/Claude API integration layer, standardized Pydantic v2 schemas, and Temporal workflow stubs — cutting new project bootstrap time from days to hours
- Built AI-powered data extraction services using OpenAI function calling and structured Pydantic output schemas to parse unstructured documents into typed, validated records for downstream ETL processing
- Implemented Temporal workflow orchestration for long-running document processing jobs, using activity retries and heartbeats to ensure reliability across multi-step pipelines handling 10,000+ documents per batch
- Built ETL pipelines for large-volume text preprocessing — including deduplication, embedding generation, and feature extraction — processing ingestion jobs that fed the vector index for semantic search features

Atrix AI (Spartan's client)

Backend Engineer | [01/2025] - [12/2025]

Developed backend infrastructure for a compliance-grade AI platform helping Medical Affairs teams in life sciences unlock insights from clinical and field data

- Built FastAPI microservices for ingesting, preprocessing, and embedding multi-source medical documents (clinical notes, field medical reports, literature) into vector databases for semantic retrieval
- Implemented RAG pipelines combining vector search (Pinecone) with metadata filtering and reranking to power accurate, citation-backed responses for HCP engagement use cases
- Integrated OpenAI GPT models and LLaMA-based models via API to generate compliant insight summaries and automated medical content drafts, enforcing prompt guardrails for regulatory safety
- Designed async background workers with FastAPI BackgroundTasks and Celery to handle long-running LLM inference jobs without blocking the main request thread
- Solved a data enrichment problem where siloed databases produced hallucination-prone LLM outputs by building a structured context layer that pre-enriches prompts with validated metadata — reducing hallucination rate by ~60% in internal evals
- Deployed services to AWS (ECS + Lambda) with GitHub Actions CI/CD pipelines and Sentry-based error monitoring for production reliability

Sidecardata.com (Spartan's client)

Backend Engineer | [03/2024] - [12/2024]

Built backend services for a consolidated DataOps platform delivering data quality, cataloging, and AI context management for lean data teams

- Designed and maintained asynchronous task pipelines using Celery and Redis to orchestrate large-scale data quality checks, catalog refreshes, and lineage updates across the warehouse

- Built RESTful API services with Django REST Framework to expose data catalog, governance rules, and observability metrics to front-end dashboards and third-party integrations
- Implemented webhook-driven event handlers integrating with Jira, Linear, and Slack to automatically trigger data pipeline workflows from ticket or message events
- Resolved a critical bottleneck in Celery Beat scheduling that caused duplicate pipeline runs under high load, reducing redundant task executions by ~40%
- Optimized Django ORM queries and introduced DB-level indexing for metadata tables, cutting average API response time from 800ms to under 200ms on catalog endpoints
- Designed background worker retry logic with exponential backoff to handle upstream warehouse timeouts, improving pipeline reliability from 87% to 99%+ success rate

DG EXTERNAL

Software Engineer | 05/2022 - 04/2023

Led development of Omni Channel ecosystem using Python/FastAPI

- Designed microservice architecture to support core business logic
- Collaborated with third-party services for seamless system integration
- Implemented network routing applications for distributed system communication
- Architected and optimized database schemas for high-performance data access
- Maintained and improved CI/CD pipelines ensuring code quality and reliability
- Conducted code reviews and mentored team members on Python best practices
- Developed comprehensive API documentation for internal and external partners

TECHNICAL SKILLS

Core Competencies

- **Backend Development:** Python (4+ years), Django, FastAPI, RESTful APIs, Celery, async programming
- **AI & LLM:** RAG pipelines (multi-source retrieval, reranking, metadata search), Vector databases (Pinecone, OpenSearch), LLM integration (OpenAI), LLM fine-tuning (LoRA, instruction tuning), LangChain
- **Database:** PostgreSQL, SQL optimization, ClickHouse, Database design, Caching (Redis)
- **Cloud & Infrastructure:** AWS (Lambda, EC2, S3, ECS), GCP, Docker, CI/CD (GitHub Actions), Linux
- **Architecture:** Microservices, Scalable systems, SaaS integration, Hybrid architecture

Additional Skills

- **Data Engineering:** Apache BEAM, ETL workflows, text preprocessing, embeddings, feature extraction
- **Monitoring & Observability:** Sentry, Grafana, logging pipelines
- **Authentication & Authorization:** OAuth2, JWT, Casbin, RBAC, PBAC
- **Frontend Knowledge:** Angular, TypeScript
- **Development Methodologies:** Scrum, Git flow

LANGUAGE PROFICIENCY

- Vietnamese: Native
- English: Intermediate